

SQL Server loves PowerShell to move Data around

Data Saturday Rheinland 2025
Andreas Jordan
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Platinum



Gold



Silver



Bronze



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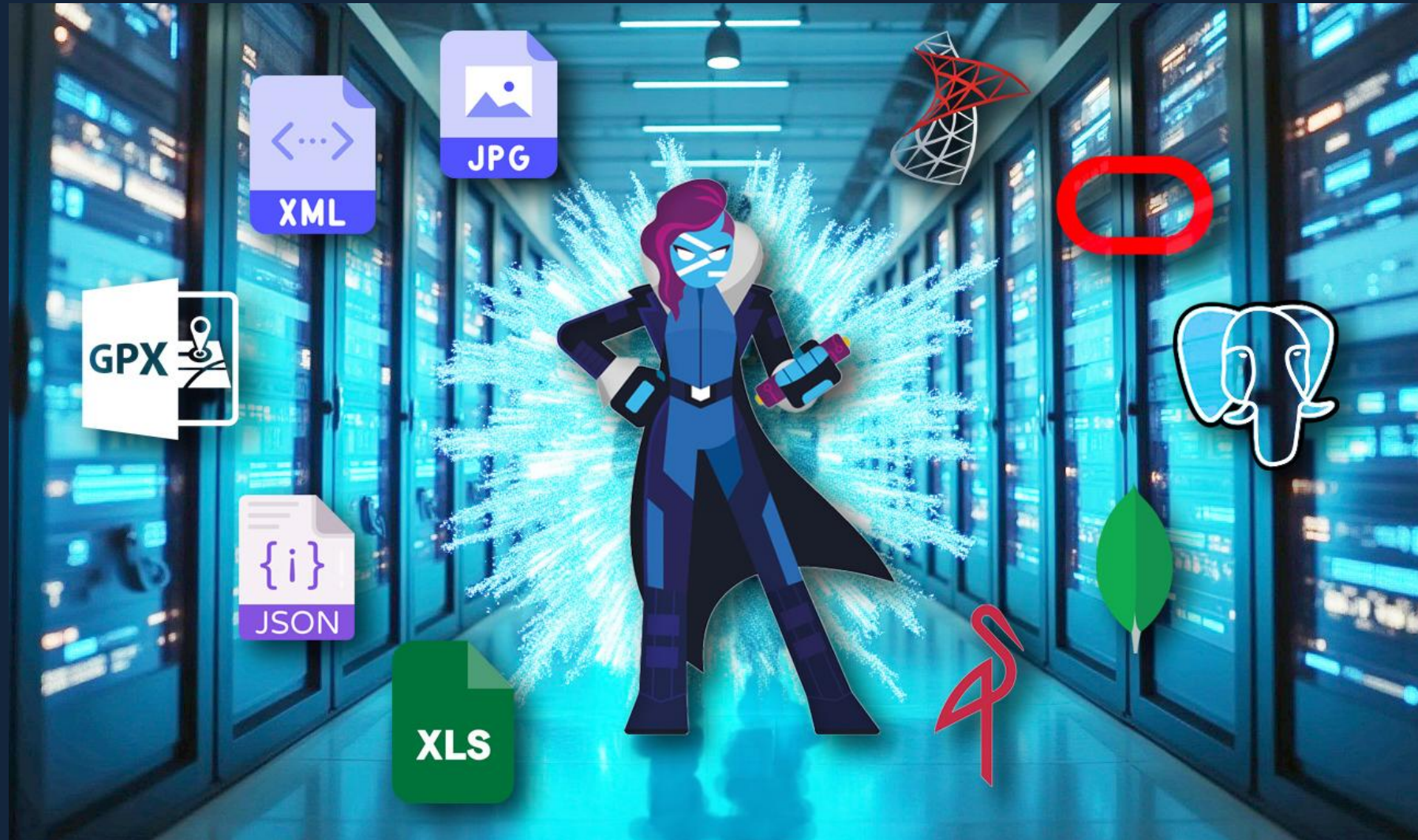
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- With ORDIX AG since 2001: Principal consultant and team lead
- Maintainer of the PowerShell module dbatools
- Technical focus
 - Microsoft SQL Server
 - Oracle Database
 - PowerShell
- <https://blog.ordix.de/andreas-jordan>



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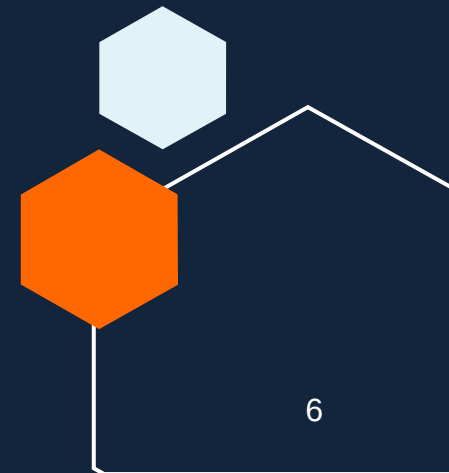


Demo Scenarios

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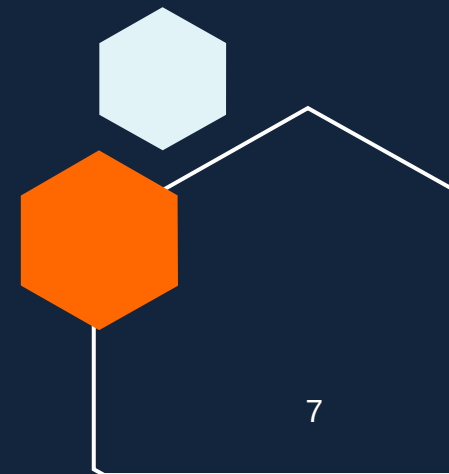
- Timesheets
 - Excel as a Database
 - The basics
- StackExchange
 - Importing XML files
 - Large Strings
 - Streaming Data around
- Geodata
 - Importing JSON files
 - Special datatypes
- PhotoService
 - Importing JPEG files
 - Changing data
 - Consistency
 - Transactions

[github.com/andreasjordan /PowerShell-moves-Data-around /tree/main/demo](https://github.com/andreasjordan/PowerShell-moves-Data-around/tree/main/demo)



Timesheets

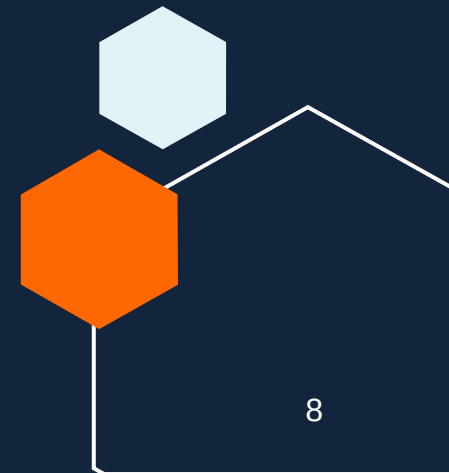
- Excel is good at storing data including datatypes, better than CSV files
- Inside of PowerShell data should be stored in arrays of PSCustomObjects
- Inserts can use BULK INSERT with some databases like SQL Server and Oracle
- If you need individual inserts, use bind parameters
- PowerShell 5.1 can still be used



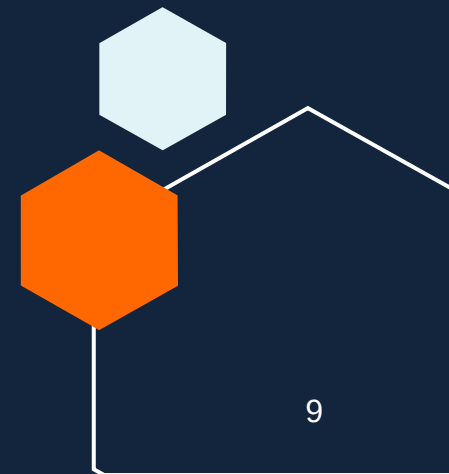
StackExchange

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- Files with a suitable format can be streamed to database tables
- .NET can take care of type conversions
- Streaming from one database to another database (system) is no problem
- Data can also be uploaded to and retrieved from NoSQL databases



- Most geodata formats can be converted into "well known text" (WKT) format
- WKT strings can be converted into database specific datatypes inside of the VALUES clause
- When selecting geodata from a table, we need to transform it back to WKT
- Two steps for more performance: BulkCopy WKT into a staging table and convert with INSERT SELECT

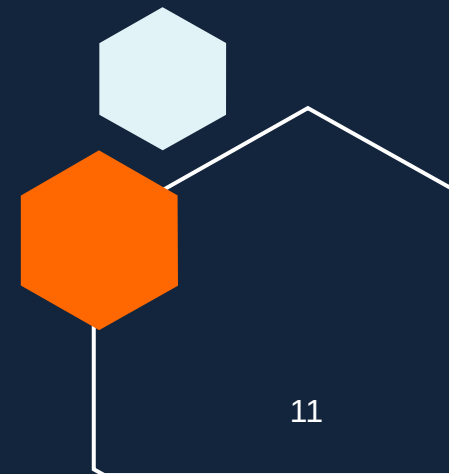


- Binary data is just data, nothing special
- Transferring only new or changed data requires some considerations
- Main question: How to identify the relevant rows
- Transactions help to get consistent data
- Using event data is sometimes an option



Key takeaways

- PowerShell loves to move Data around
- PowerShell supports Data driven development
- Wrapper commands can create a consistent way of interaction with different data sources



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**Many thanks
for your attention**



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